

SHARK TANK FINAL PROJECT – INDIVIDUAL VERSION

COURSE: Fundamentals of Web Development (Hoar, 3rd Edition)

FORMAT: Individual Project

DELIVERABLE: Working Web Application + Pitch + Documentation

PROJECT OVERVIEW

You will create a brand-new product, service, concept, or technology and build a functional web-based prototype for it. You will then pitch your idea in a Shark Tank-style presentation. This is an individual project that integrates all major topics from the course, including HTML, CSS, JavaScript, React, Node.js, databases, state management, security, and deployment.

Your product can be anything you imagine: a tool, an app, a service, a device interface, a social platform, a productivity system, a game, or a new technology concept. The key requirement is that your idea must be supported by a working full-stack web application.

UPDATED TECHNICAL REQUIREMENTS

1. HTML Requirements

- Semantic HTML5 structure using header, nav, main, section, article, footer.
- One interactive form with text input, dropdown, radio/checkboxes, submit button, and validation.
- One table or structured data display.
- Two types of web media: images, video, audio, or canvas.

2. CSS Requirements

- Use of class, ID, and attribute selectors.
- Responsive layout using Flexbox or Grid with media queries.
- Consistent design system: color palette, typography hierarchy, spacing scale.
- Styled UI components: buttons, cards, nav bar, forms.

3. JavaScript Requirements

- Use of variables, arrays, objects, loops, and functions.
- DOM manipulation for dynamic rendering and UI updates.
- ES modules for code organization.
- React components with useState, props, and conditional rendering.

- Event handling for forms, buttons, sorting, filtering.

4. Node.js Server Requirements

- At least three API routes: GET, POST, PUT/PATCH, DELETE.
- Data validation and transformation.
- Environment variables via .env for credentials and keys.
- Modular structure with routes, controllers, and utilities.

5. Database Requirements

- Use MongoDB, PostgreSQL, MySQL, or SQLite.
- At least two collections/tables.
- Full CRUD operations.
- Clear schema or model definitions.

6. State Management Requirements

- Component-level state with useState.
- One example of global state using Context API.
- Persistent state using localStorage or sessionStorage.

7. Security Requirements

- Input sanitization on client and server.
- Secure API key handling with .env.
- Basic authentication with registration, login, hashed passwords, and protected routes.

8. DevOps & Hosting Requirements

- Deploy front-end on Netlify or Vercel.
- Deploy back-end on Render or Railway.
- GitHub repo with commit history.
- Use analytics tool such as Google Analytics or Vercel Analytics.

FINAL DELIVERABLES

- 1. Working Web Application
- 2. Shark Tank Pitch (5–7 minutes)
- 3. Documentation Packet (README, API docs, ERD, screenshots, slides)
- 4. Testing (5 JS unit tests, 2 API tests)

GRADING BREAKDOWN (100 POINTS)

- Front-End: 20
- Back-End: 15
- Database: 10
- State Management: 10

- Security: 5
- DevOps & Hosting: 10
- Testing: 10
- Pitch: 15
- Documentation & Code Quality: 5

PROJECT GUIDELINES

Your product must solve a real problem, include a clear user flow, demonstrate technical depth, show creativity, and be built entirely by you.

SUGGESTED PRODUCT IDEAS

- Smart fitness dashboard
- AI-powered study assistant
- Eco-friendly shopping tool
- Micro-community social platform
- Habit-tracking wellness app
- Music-driven mood analyzer
- Productivity automation tool
- Game-based learning system

SUBMISSION INSTRUCTIONS

Submit a ZIP file named: SharkTank_FinalProject_YourLastName.zip including all source code, documentation, slides, tests, and README with deployment link.